

Applied Optimization

Solution to Exercise 2 - Convex Functions

Convex Functions (1 pt)

Simple functions (0.5 pt)

Are the following functions convex? Prove your statement.

1. $f(x) = x^2, x \in \mathbb{R}$
2. $f(x) = e^{x^2}, x \in \mathbb{R}$
3. $f(x, y) = x^2 + 3xy + 2y^2, x \in \mathbb{R}, y \in \mathbb{R}$

Solution:

1. $f'(x) = 2x, f''(x) = 2$. Since the second derivative is larger than 0, it is convex.
2. $f'(x) = 2xe^{x^2}, f''(x) = 2e^{x^2} + 4x^2e^{x^2}$. The second derivative is always larger than 0 in $\mathbb{R}$. So it is convex.
3. $f'(x, y) = (2x + 3y, 4y + 3x)^T, f''(x, y) = \begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix}$. The eigenvalues of the hessian matrix are $3 + \sqrt{10}$ and $3 - \sqrt{10}$. The function is not convex because $3 - \sqrt{10} < 0$.

Log-sum-exp (0.5 pt)

Show that the function

$$f(x) = \log(e^{x_1} + \dots + e^{x_n})$$

is convex on $\mathbb{R}^n$.

Hint: The proof is outlined in the Boyd's book p. 74. Develop it by yourself and provide the intermediate steps.

Solution:

Let $z = (e^{x_1}, \dots, e^{x_n})$ and $\mathbf{1}$ be a vector in $\mathbb{R}^n$ whose entries are all ones. Then this f can be rewritten as:

$$f(x) = \log(\mathbf{1}^T z)$$

now compute the gradient and hessian of f , by applying the chain rule we get

$$\begin{aligned}\frac{\partial}{\partial x_i} \log(\mathbf{1}^T z) &= \frac{1}{\mathbf{1}^T z} \cdot \frac{\partial}{\partial z_i} (\mathbf{1}^T z) \cdot \frac{\partial z}{\partial x_i} \\ &= \frac{1}{\mathbf{1}^T z} \cdot 1 \cdot e^{x_i}\end{aligned}$$

thus

$$\begin{aligned}\nabla f(x) &= \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right) \\ &= \left(\frac{1}{\mathbf{1}^T z} \cdot e^{x_1}, \dots, \frac{1}{\mathbf{1}^T z} \cdot e^{x_n} \right) \\ &= \frac{1}{\mathbf{1}^T z} (e^{x_1}, \dots, e^{x_n}) \\ &= \frac{1}{\mathbf{1}^T z} z\end{aligned}$$

Now for the Hessian matrix, each element is defined as:

$$\begin{aligned}\mathbf{H}_{i,j} &= \frac{\partial^2 f}{\partial x_j \partial x_i} \\ &= \frac{\partial}{\partial x_j} \frac{\partial}{\partial x_i} \log(\mathbf{1}^T z) \\ &= \frac{\partial}{\partial x_j} \frac{e^{x_i}}{\mathbf{1}^T z} \\ &= \delta_{i,j} e^{x_i} \cdot \frac{1}{\mathbf{1}^T z} - \frac{1}{(\mathbf{1}^T z)^2} \cdot \frac{\partial z}{\partial x_j} \cdot e^{x_i} \\ &= \delta_{i,j} e^{x_i} \cdot \frac{1}{\mathbf{1}^T z} - \frac{1}{(\mathbf{1}^T z)^2} \cdot e^{x_j} \cdot e^{x_i}\end{aligned}$$

and thus

$$\mathbf{H} = \nabla^2 f(x) = \frac{1}{\mathbf{1}^T z} \text{diag}(z) - \frac{1}{(\mathbf{1}^T z)^2} z z^T$$

Now let us show that

$$v^T \nabla^2 f(x) v = \frac{1}{\mathbf{1}^T z} v^T \text{diag}(z) v - \frac{1}{(\mathbf{1}^T z)^2} v^T z z^T v \geq 0$$

By applying the suggested change of variables:

$$\begin{aligned} a &= (v_1 \sqrt{z_1} \dots v_n \sqrt{z_n})^T, \quad b = (\sqrt{z_1} \dots \sqrt{z_n})^T \\ a^T a &= (v_1 \sqrt{z_1} \dots v_n \sqrt{z_n}) \begin{pmatrix} v_1 \sqrt{z_1} \\ \vdots \\ v_n \sqrt{z_n} \end{pmatrix} = v_1^2 z_1 + \dots + v_n^2 z_n \\ b^T b &= (\sqrt{z_1} \dots \sqrt{z_n}) \begin{pmatrix} \sqrt{z_1} \\ \vdots \\ \sqrt{z_n} \end{pmatrix} = z_1 + \dots + z_n = \mathbf{1}^T z \\ a^T b &= (v_1 \sqrt{z_1} \dots v_n \sqrt{z_n}) \begin{pmatrix} \sqrt{z_1} \\ \vdots \\ \sqrt{z_n} \end{pmatrix} = v_1 z_1 + \dots + v_n z_n = v^T z = z^T v \end{aligned}$$

$$\begin{aligned} v^T \nabla^2 f(x) v &= \frac{1}{\mathbf{1}^T z} v^T \text{diag}(z) v - \frac{1}{(\mathbf{1}^T z)^2} v^T z z^T v \\ &= \frac{1}{b^T b} v^T \begin{pmatrix} z_1 v_1 \\ \vdots \\ z_n v_n \end{pmatrix} - \frac{1}{(b^T b)^2} (a^T b)(a^T b) \\ &= \frac{1}{b^T b} a^T a - \frac{1}{(b^T b)^2} (a^T b)^2 \end{aligned}$$

By using the *Cauchy-Schwarz* inequality, valid for any two vectors a and b

$$\begin{aligned} (a^T a)(b^T b) &\geq (a^T b)^2 \\ (a^T a)(b^T b) - (a^T b)^2 &\geq 0, \quad \text{divide by } (b^T b)^2 > 0 \\ \frac{1}{b^T b} a^T a - \frac{1}{(b^T b)^2} (a^T b)^2 &\geq 0 \end{aligned}$$

we can conclude that $\nabla^2 f(x)$ is positive semi-definite, thus f is convex by the second-order condition.

Bonus (0.5 pt)

Geometric mean

Show that the geometric mean

$$f(\mathbf{x}) = (\prod_{i=1}^n x_i)^{1/n}$$

is concave on $\mathbb{R}_{++}^n$.

Solution:

First start by verifying the given expressions for the Hessian

$$\frac{\partial^2 f(x)}{\partial x_k^2} = -(n-1) \frac{(\prod_{i=1}^n x_i)^{1/n}}{n^2 x_k^2}, \quad \frac{\partial^2 f(x)}{\partial x_k \partial x_l} = \frac{(\prod_{i=1}^n x_i)^{1/n}}{n^2 x_k x_l} \quad k \neq l,$$

for which we need to compute the gradient. Note that the derivative of the product $\prod_{i=1}^n x_i$ with respect to x_k is another product with all variables except x_k , $\prod_{i=1}^{k-1} x_i \prod_{i=k+1}^n x_i$

$$\begin{aligned} \frac{\partial f(x)}{\partial x_k} &= \frac{1}{n} \left(\prod_{i=1}^n x_i \right)^{1/n-1} \left(\prod_{i=1}^{k-1} x_i \prod_{i=k+1}^n x_i \right) \\ &= \frac{1}{n} \left(\prod_{i=1}^n x_i \right)^{1/n} \frac{\prod_{i=1}^{k-1} x_i \prod_{i=k+1}^n x_i}{\prod_{i=1}^n x_i} \\ &= \frac{1}{n} \left(\prod_{i=1}^n x_i \right)^{1/n} \frac{1}{x_k}. \end{aligned}$$

The hessian is obtained via deriving each k -th entry of the gradient by x_l for $k, l = 1, \dots, n$. Specifically, when $l = k$

$$\begin{aligned} \frac{\partial^2 f(x)}{\partial x_k^2} &= \frac{1}{n} \frac{\partial}{\partial x_k} \left[\left(\prod_{i=1}^n x_i \right)^{1/n} \right] \frac{1}{x_k} + \frac{1}{n} \left(\prod_{i=1}^n x_i \right)^{1/n} \frac{\partial}{\partial x_k} \left(\frac{1}{x_k} \right) \\ &= \frac{1}{n} \left[\frac{1}{n} \left(\prod_{i=1}^n x_i \right)^{1/n} \frac{1}{x_k} \right] \frac{1}{x_k} + \frac{1}{n} \left(\prod_{i=1}^n x_i \right)^{1/n} \left(-\frac{1}{x_k^2} \right) \\ &= \frac{(\prod_{i=1}^n x_i)^{1/n} - n (\prod_{i=1}^n x_i)^{1/n}}{n^2 x_k^2} \\ &= \frac{-(n-1)}{n^2 x_k^2} \left(\prod_{i=1}^n x_i \right)^{1/n} \end{aligned}$$

give us the expression for $\frac{\partial^2 f(x)}{\partial x_k^2}$. Similarly deriving with respect to $x_l, l \neq k$

$$\begin{aligned}\frac{\partial^2 f(x)}{\partial x_k \partial x_l} &= \frac{1}{n} \frac{\partial}{\partial x_l} \left[\left(\prod_{i=1}^n x_i \right)^{1/n} \right] \frac{1}{x_k} \\ &= \frac{1}{n} \left[\frac{1}{n} \left(\prod_{i=1}^n x_i \right)^{1/n} \frac{1}{x_l} \right] \frac{1}{x_k} \\ &= \frac{1}{n^2 x_k x_l} \left(\prod_{i=1}^n x_i \right)^{1/n}\end{aligned}$$

give us the expression for $\frac{\partial^2 f(x)}{\partial x_k \partial x_l}$ for $l \neq k$.

Second, to write the hessian in matrix notation we introduce vector $q = \left(\frac{1}{x_1}, \dots, \frac{1}{x_n} \right)^T$. That is $q_i = \frac{1}{x_i}$. Note that the product qq^T produces a matrix with entries $\frac{1}{x_k x_l}$. Thus

$$\frac{\partial^2 f(x)}{\partial x_k \partial x_l} = \frac{1}{n^2} \left(\prod_{i=1}^n x_i \right)^{1/n} qq^T.$$

Note however that the above matrix does not reproduce $\frac{\partial^2 f(x)}{\partial x_k^2}$ when $k = l$,

$$\left. \frac{\partial^2 f(x)}{\partial x_k \partial x_l} \right|_{l=k} = \frac{1}{n^2 x_k^2} \left(\prod_{i=1}^n x_i \right)^{1/n}$$

the missing term is $-\frac{1}{n x_k^2} \left(\prod_{i=1}^n x_i \right)^{1/n}$, thus we add it the diagonal matrix $\frac{-1}{n} \left(\prod_{i=1}^n x_i \right)^{1/n} \text{diag} \left(\frac{1}{x_1^2}, \dots, \frac{1}{x_n^2} \right)$ to get the complete hessian expression

$$\nabla^2 f(x) = -\frac{1}{n^2} \left(\prod_{i=1}^n x_i \right)^{1/n} \left(n \text{diag} \left(\frac{1}{x_1^2}, \dots, \frac{1}{x_n^2} \right) + qq^T \right)$$

Finally, to show that $f(x)$ is convex on $\text{dom} f$ we use the second-order condition. We show that the matrix $\nabla^2 f(x)$ is negative semi-definite, that is

$$v^T \nabla^2 f(x) v \leq 0$$

for any $v \in \mathbb{R}^n$ and any $x \in \text{dom} f = \mathbb{R}_{++}^n$. To do that we make use of the Cauchy-Schwarz inequality with the following substitutions: $a_i = 1$, a is a vector of n ones, and $b_i = v_i / x_i$.

$$a^T a = n, \quad a^T b = b^T a = \sum_{i=1}^n v_i / x_i, \quad b^T b = \sum_{i=1}^n v_i^2 / x_i^2$$

$$\begin{aligned}
v^T \nabla^2 f(x) v &= -\frac{1}{n^2} \left(\prod_{i=1}^n x_i \right)^{1/n} \left(\underbrace{n}_{=a^T a} v^T \text{diag} \left(\frac{1}{x_1^2}, \dots, \frac{1}{x_n^2} \right) v + \underbrace{v^T q}_{(v_1, \dots, v_n)^T} \underbrace{q^T v}_{\begin{pmatrix} 1/x_1 \\ \vdots \\ 1/x_n \end{pmatrix}} \right) \\
&= -\underbrace{\frac{1}{n^2} \left(\prod_{i=1}^n x_i \right)^{1/n}}_{>0} \underbrace{\left((a^T a)(b^T b) - (a^T b)^2 \right)}_{\geq 0}
\end{aligned}$$

Thus, because of the leading negative sign and the two terms being non-negative, we conclude $v^T \nabla^2 f(x) v \leq 0$ and $f(x)$ is concave in $\mathbb{R}_{++}^n$.